FINAL CALIBRATION AND PROCESS PERFORMANCE REPORT

LAM Research Challenge 2026 | Stage 3 | Day 1 and Day 2

Evidence base: 16 logged trials in calibration_trials.csv, interpreted alongside the supplied calibration narrative. Values below distinguish measured outcomes from estimates and narrative-only claims.

16 logged trials	**4** verified K-factors	**99.90%** flush accuracy	**+1.67%** estimated net batch error

Executive findings

The calibration sequence corrected large initial under-delivery and produced close final volume agreement in the logged verification runs. The three colour-channel checks (Trials 10-12) had absolute errors of 0.06%-0.13%; the dedicated flush-water verification (Trial 15) was 2 mL below its 2,000 mL target. These are measured water checks; the dataset does not provide independent paint-only calibration runs.

The final mixer batch recorded 3,150 mL against a 3,000 mL recipe target (+5.00% gross). The report attributes about 100 mL to water displaced from line hold-up, which yields an estimated 3,050 mL paint dose and +1.67% net error. The source log does not measure that water/paint split directly, so the net result is an estimate rather than a directly observed measurement.

Performance at a glance

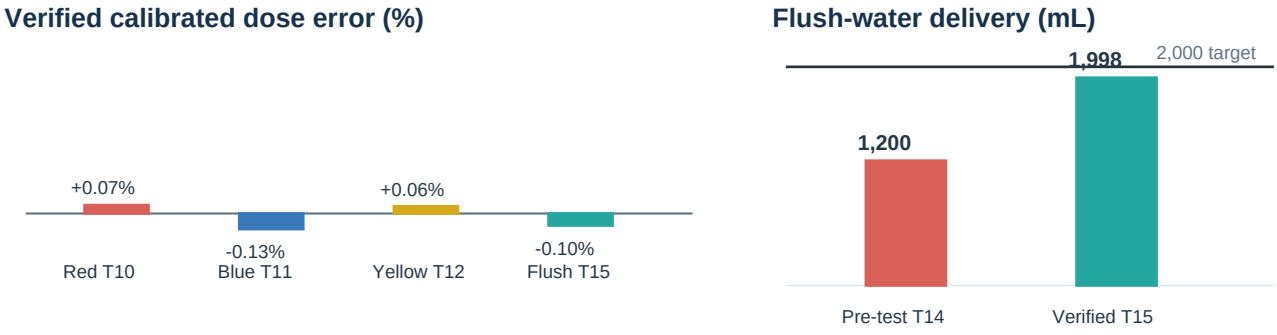


Figure 1. Logged calibrated volume errors and the flush-water correction. Trial 14 used the blue-channel constant on water and delivered 1,200 mL; Trial 15 used the dedicated flush constant and delivered 1,998 mL.

Overall assessment

Calibration outcome: Successful for the measured water runs. **Flush outcome:** 99.90% of target by physical volume. **Batch outcome:** gross volume above target; net paint error is +1.67% only under the approximately 100 mL line-water assumption. **Automation outcome:** the supplied narrative reports automatic step transitions and safe shutdown, but the CSV alone does not log those events or a temperature trace.

1. Calibration results

K-factor is calculated as counted pulses divided by measured volume (pulses/mL). The final colour-channel constants were derived from water dosing trials and should be interpreted as calibrated values for those logged conditions.

Stream / trial	Pulses	Target	Measured	K-factor	Error
Red / T10	196	300 mL	300.2 mL	0.654 p/mL	+0.07%
Blue / T11	310	300 mL	299.6 mL	1.034 p/mL	-0.13%
Yellow / T12	4,992	2,400 mL	2,401.5 mL	2.080 p/mL	+0.06%
Flush water / T15	3,443	2,000 mL	1,998.0 mL	1.723 p/mL	-0.10%

Calibration progression

Stage	Evidence	Interpretation
Initial characterization (T1-6)	50-100 mL timed runs; 10 mL/s reported across channels.	Baseline pulse counts were near 0.65, 1.03-1.04 and 2.08 p/mL.
Initial closed-loop dose (T7-9)	All three channels delivered half of target.	K values were effectively doubled in the first dose configuration; recalibration was required.
Verified channel doses (T10-12)	300.2 / 299.6 / 2,401.5 mL vs 300 / 300 / 2,400 mL.	Absolute errors 0.07%, 0.13%, and 0.06%.
Flush water correction (T14-15)	1,200 mL at K=1.034; then 1,998 mL at K=1.723.	Dedicated water constant resolved the 40% under-delivery in the pre-test.

Flush calculation

Trial 14 measured 1,200 mL after 2,068 pulses while using 1.034 p/mL. The derived water constant is 2,068 / 1,200 = 1.723 p/mL. Trial 15 then delivered 1,998 mL from 3,443 pulses; measured error = (1,998 - 2,000) / 2,000 = -0.10%, or -2.0 mL.

The data support a fluid-specific calibration difference for FT-03 under these test conditions. The draft narrative attributes this to viscosity; viscosity is a plausible explanation, but the trials do not isolate viscosity from other setup or operating differences.

2. Batch performance and interpretation

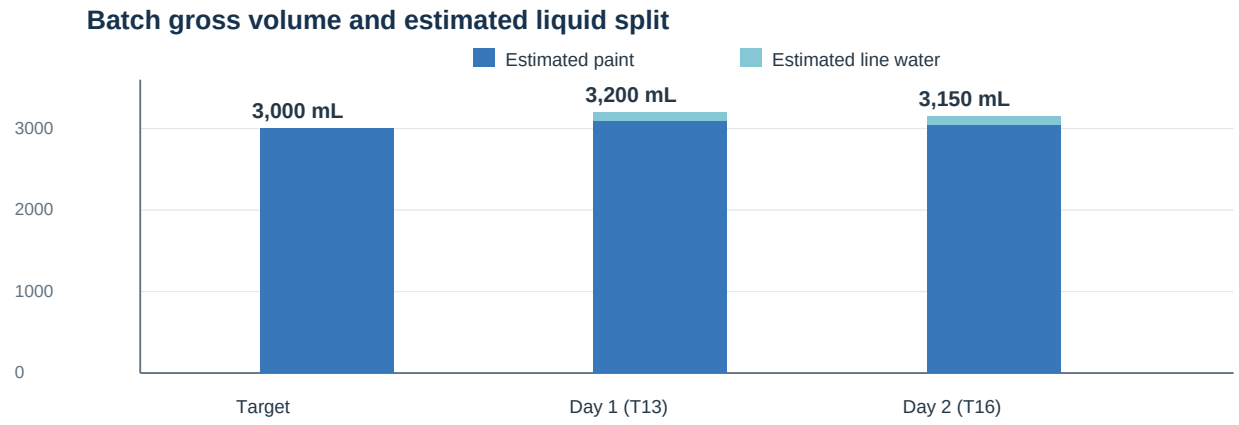
The batch trials record total collected liquid, not separately measured paint and water fractions. The table therefore reports gross volume as observed and net paint volume only as an estimate based on the source narrative's approximately 100 mL line-water displacement assumption.

Metric	Target	Day 1 initial (T13)	Day 2 demo (T16)
Gross liquid collected	3,000 mL	3,200 mL (+6.67%)	3,150 mL (+5.00%)
Estimated line water	0 mL	~100 mL*	~100 mL*
Estimated paint delivered	3,000 mL	~3,100 mL	~3,050 mL
Estimated net paint error	0.00%	+3.33% (+100 mL)	+1.67% (+50 mL)
Pulse total	-	5,498	5,498
Batch note	Mustard recipe	Mustard achieved (log)	Mustard confirmed (log)

*Line-water volume is an estimate from the supplied narrative, not a channel measurement in the CSV. The draft estimates 105.6 mL from three 70 cm, 8 mm-ID tubes, then rounds this to about 100 mL for batch decomposition. The paint-only delivery volumes and net errors inherit that assumption.

Batch result

The Day 2 batch collected 150 mL more than the 3,000 mL total target. If about 100 mL was displaced line water, the estimated paint portion is 3,050 mL, 50 mL above target. The available log does not identify per-colour delivered volumes in either batch, so it cannot establish which colour channel contributed the overshoot or verify the 80/10/10 blend in the batch itself.



Gross batch totals are measured; the approximately 100 mL water split is an estimate from the supplied narrative.

3. Complete trial record

Values are transcribed from data/calibration_trials.csv. K-factor is shown as reported in that file; "—" indicates zero/not applicable for combined batch trials.

#	Day	Fluid	Line	Test type	s	Pulses	Target mL	Actual mL	K p/mL	Error
1	1	WATER	RED	timed burst	5	33	50	50.0	0.660	+0.00 %
2	1	WATER	RED	steady run	10	65	100	100.0	0.650	+0.00 %
3	1	WATER	BLUE	timed burst	5	52	50	50.0	1.040	+0.00 %
4	1	WATER	BLUE	steady run	10	103	100	100.0	1.030	+0.00 %
5	1	WATER	YELLOW	timed burst	5	104	50	50.0	2.080	+0.00 %
6	1	WATER	YELLOW	steady run	10	208	100	100.0	2.080	+0.00 %
7	1	WATER	YELLOW	closed loop dose	120	2,496	2,400	1,200.0	1.040	-50.00 %
8	1	WATER	RED	closed loop dose	15	98	300	150.0	0.327	-50.00 %
9	1	WATER	BLUE	closed loop dose	15	155	300	150.0	0.517	-50.00 %
10	1	WATER	RED	calibrated dose	30	196	300	300.2	0.654	+0.07 %
11	1	WATER	BLUE	calibrated dose	30	310	300	299.6	1.034	-0.13%
12	1	WATER	YELLOW	calibrated dose	240	4,992	2,400	2,401.5	2.080	+0.06 %
13	1	Paint mix	ALL	initial batch	240	5,498	3,000	3,200.0	—	+6.67 %
14	2	WATER	FLUSH	step1 uncal	120	2,068	2,000	1,200.0	1.034	-40.00 %
15	2	WATER	FLUSH	step1 verified	200	3,443	2,000	1,998.0	1.723	-0.10%
16	2	Paint mix	ALL	demo batch	240	5,498	3,000	3,150.0	—	+5.00 %

Calculation notes

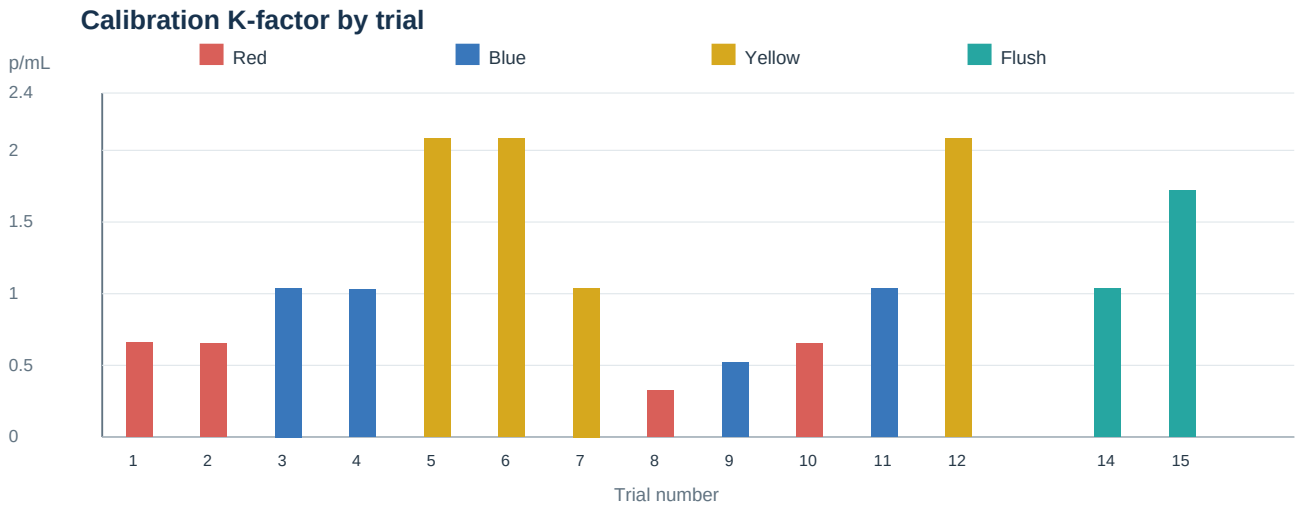
Volume error (%) = (actual volume - target volume) / target volume × 100. Accuracy is reported as 100% minus absolute percent error. K-factor = pulses / measured volume. For trials 13 and 16, CSV reports K=0.000 and combined batch volume; this report does not infer individual channel calibration from those totals.

Source files

Primary measurement source: data/calibration_trials.csv (16 rows). Context source: CALIBRATION_REPORT.md, used for reported architecture, recipe and the estimated line-water explanation. Claims absent from the CSV are identified as narrative-reported or estimated in this report.

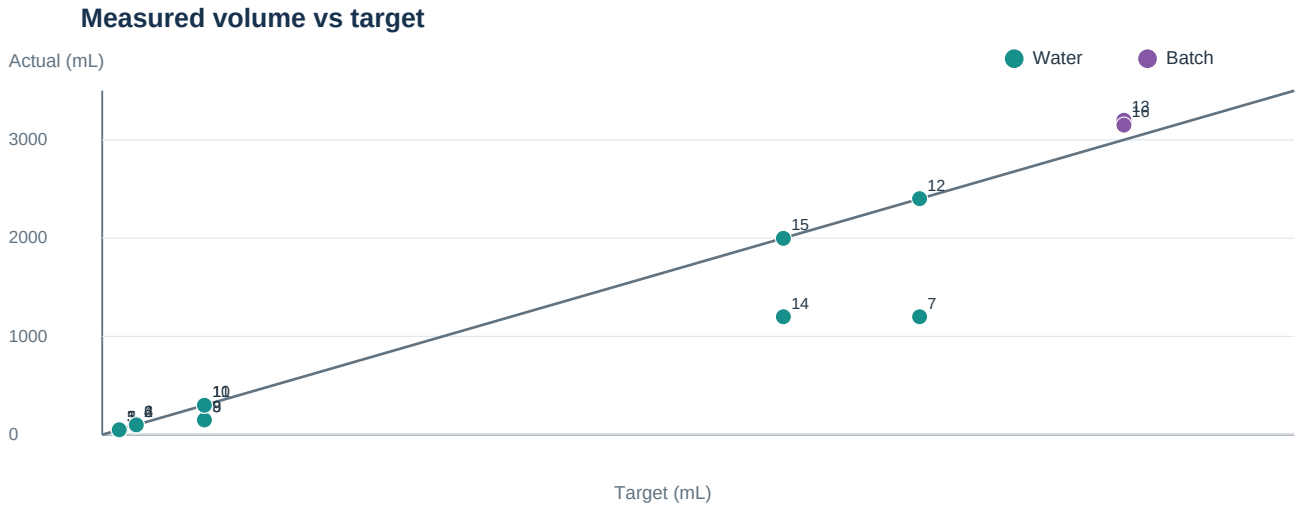
4. Plots from the logged trials

These plots use measured values in the CSV. Trial 13 and 16 combined-batch K-factors are zero/not applicable and are omitted from the K-factor chart.

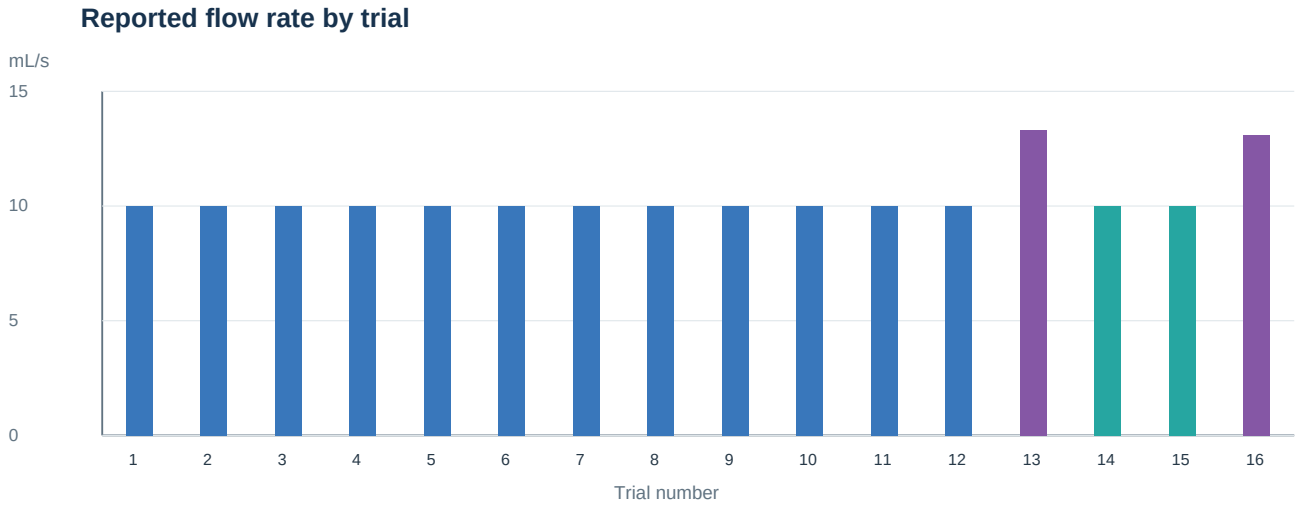


Trials 7-9 show the early under-delivery configuration; Trial 14 is the flush pre-test using the blue constant. Trials 10-12 and 15 show the verified stream-specific settings.

4. Plots from the logged trials (continued)



Dashed line marks exact target. Numbers label trial IDs. Early closed-loop doses (7-9) sit well below the line; batch trials sit above it.



Flow rates are the CSV-reported values. Most water trials report 10.0 mL/s; combined batch trials report 13.3 and 13.1 mL/s.

Plot interpretation

The most visible calibration issue is the approximately 50% under-delivery in Trials 7-9. The dedicated flush constant corrected the pre-test gap from -40.00% to -0.10%. Batch gross totals remained above recipe target; net paint error uses the line-water estimate described on page 3.